

Audio Encoders and Speech Models

DATA 443/643 · Week 6, note 6.2

Week 6

American University

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Ideas from Stanford CS224S 2025 L11 (Whisper: log-mel, two convs, GPT-style decoder, 30 s, no CTC). Original slides; that course is not copied.

Lecture plan

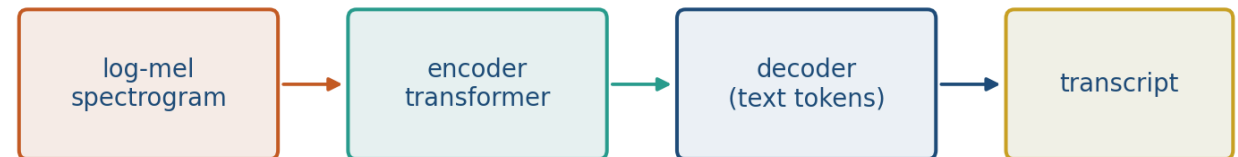
- 1 Whisper: encoder then a causal decoder
- 2 30 seconds is 3000 frames, not 480k samples
- 3 Special tokens: language, task, time
- 4 CLAP: CLIP for audio

Key goal: pick Whisper vs CLAP by the head, not by the spectrogram.

Whisper is translation, with a spectrogram

- Log-mel in. Two convs. Transformer encoder.
- Decoder is GPT-style and causal.
CS224S: no CTC.
- Cross-attention into encoder states.
Same as note 3.3.

Whisper-style: audio encoder, text decoder (Week 3 block, new input)



Why the spectrogram exists

$$30 \text{ s} \times 16 \text{ kHz} = 480,000 \quad \text{vs} \quad 30 / 0.010 = 3000$$

3000 mel frames, 80 bands, then a stride-2 stem.

About 1500 encoder steps, not a 480k-token transformer.

Long audio: 30 s windows. Timestamp tokens shift the window.

Thirty seconds, then tokens

- Special tokens: language, `<|transcribe|>`, timestamps.
- The loss is next-token on the text, conditioned on audio.
- Timestamps are vocabulary, not a second model.

The spectrogram exists so the encoder is not a 480k-token transformer.

Raw waveform

30 s at 16 kHz
= 480,000 samples

Whisper front end (CS224S L11)

10 ms hop → 3000 mel frames

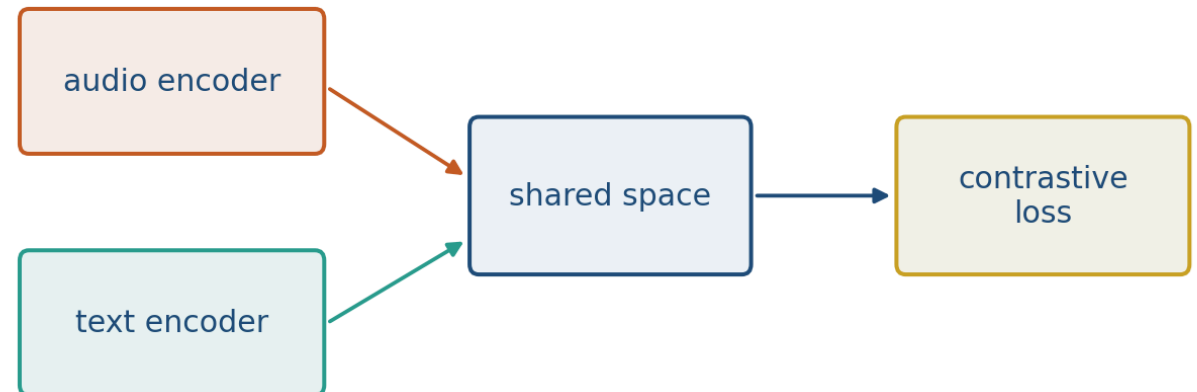


stride-2 conv → ~1500 tokens

CLAP is CLIP for sound

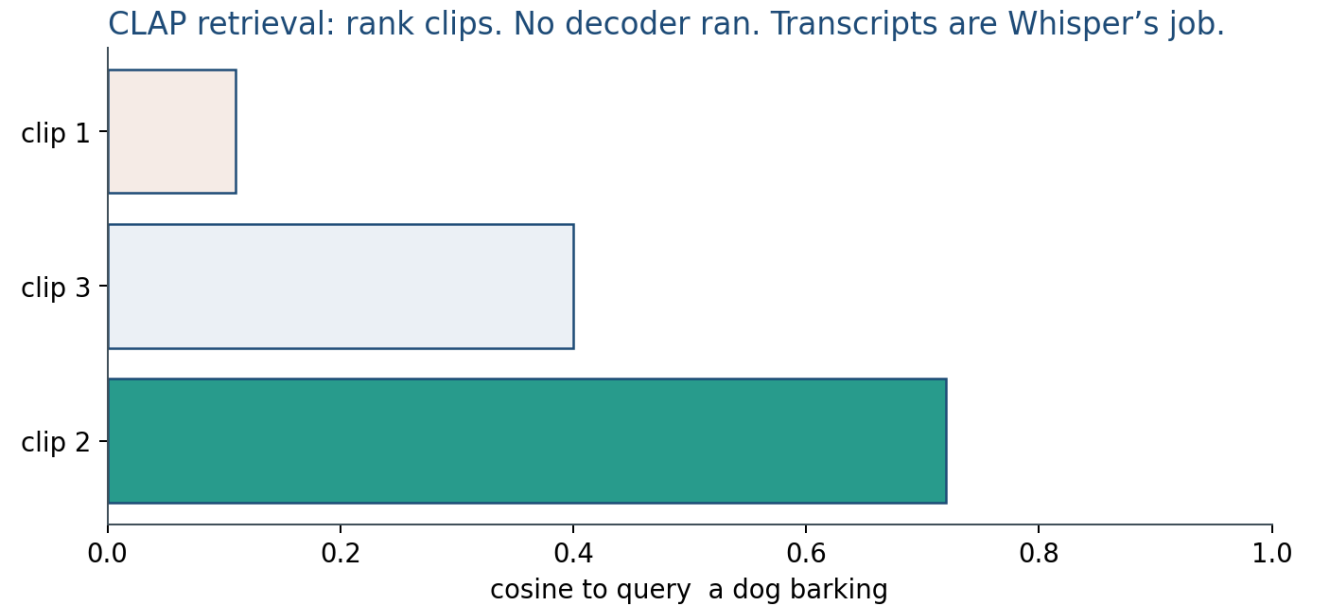
- Audio tower and text tower. InfoNCE on captions.
- Rank clips by cosine. Do not decode a transcript.
- The text tower may be bidirectional. Whisper's decoder may not.

CLAP is CLIP for audio: two towers, cosine, paired captions



Rank, then stop

- Cosines 0.11, 0.72, 0.40. Rank is clip 2, 3, 1.
- If you needed the words in clip 2, call Whisper on the wave.
- Same spectrogram tokens. Different head.



What to take from this lecture

Transcript

Whisper. Subtitles and timestamps.

Search

CLAP. Zero-shot tags. Dual retrieval.

Project

Do not concat Whisper states into CLIP without naming the rates.